Macroscopic L^∞ bounds for the Landau–Coulomb equation

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Abstract

We prove positive-time L^∞ bounds for smooth rapidly decaying solutions of the spatially homogeneous Landau equation with Coulomb interaction in three dimensions. After fixing mass, momentum and energy, the bound on each positive waiting interval depends only on an upper bound for the initial entropy and on the time interval. It is independent of initial higher moments, Fisher information and pointwise bounds. A weighted Fisher estimate first gives concentration control compatible with the second moment. To control large velocities, we use two diffusion directions supplied by a bounded part of the distribution together with the full nonlinear collision energy. A tangential kernel identity yields a nonlinear Nash inequality, from which a local small-entropy argument excludes escaping density peaks. A local parabolic estimate then controls the bounded velocity region.

1 Introduction

The homogeneous Landau equation with Coulomb interaction describes collisions between charged particles in a spatially uniform plasma. For a nonnegative velocity density $f(t,v)$ on $\mathbb{R}^3$, we use the normalization

$$\partial_t f = \nabla_v \cdot (A[f]\nabla_v f - f\nabla_v a[f]), \tag{1.1}$$

where

$$A[f](v) = \frac{1}{8\pi}\int_{\mathbb{R}^3} \frac{\Pi(v-w)}{|v-w|} f(w)\,\mathrm{d}w, \qquad a[f](v) = \frac{1}{4\pi}\int_{\mathbb{R}^3} \frac{f(w)}{|v-w|}\,\mathrm{d}w, \qquad \Pi(z) = I_3 - \frac{z\otimes z}{|z|^2}. \tag{1.2}$$

The projection is defined for $z \neq 0$; its value at zero does not affect the locally integrable convolution. Mass, momentum and energy are conserved, and the Boltzmann entropy $\int f\log f$ decreases. The regularizing effect of this nonlocal diffusion is delicate: it competes with the quadratic reaction in the equivalent equation $f_t = A[f] : D^2 f + f^2$, while its ellipticity deteriorates at large velocities.

The global classical theory changed with the Fisher information monotonicity of Guillen and Silvestre [5]. For initial data with a Maxwellian upper bound they constructed a global classical solution; the ordinary Fisher information controls its L^3 norm throughout the evolution. Golding, Gualdani and Loher [3, Theorem 1.1] established a local theory for polynomially weighted $L^{3/2}$ data, with quantitative short-time estimates. Their global theorem [3, Theorem 1.3 and (1.5)] additionally assumes a higher moment and a compatible polynomial lower bound on the initial density. Fisher information production was also proved by Ji [6, Theorem 1.4] and by Desvillettes, Golding, Gualdani and Loher [1, Theorem 1]; in the Coulomb case the respective constants retain a fifth and an eighteenth initial moment. These results distinguish global smoothness for each datum from a bound uniform over a class specified only by macroscopic quantities.

The first open problem in [3, Section 1] asks whether smooth solutions admit regularization estimates using only mass, energy and entropy, and then asks for the conjectured $1/t$ rate. We prove

the finite-bound statement for normalized smooth rapidly decaying solutions. The normalization is the same as in that paper:

$$\int_{\mathbb{R}^3} f(0,v)\,\mathrm{d}v = 1, \qquad \int_{\mathbb{R}^3} vf(0,v)\,\mathrm{d}v = 0, \qquad \int_{\mathbb{R}^3} |v|^2 f(0,v)\,\mathrm{d}v = 3. \tag{1.3}$$

Theorem 1.1. *For every $H \in \mathbb{R}$ and every $0 < \tau < T < \infty$, there is a finite constant $C(H,\tau,T)$ with the following property. Every nonnegative smooth rapidly decaying solution of (1.1) on $[0,T]\times\mathbb{R}^3$ satisfying (1.3) and*

$$\int_{\mathbb{R}^3} f(0,v)\log f(0,v)\,\mathrm{d}v \le H$$

obeys

$$\sup_{\tau\le t\le T} \|f(t)\|_{L^\infty(\mathbb{R}^3)} \le C(H,\tau,T). \tag{1.4}$$

The constant is independent of all initial higher moments, Fisher information, pointwise bounds and derivative norms.

The theorem concerns the macroscopic dependence of an a priori bound on a given smooth solution, including nonnegative initial profiles with vacuum. No sharp waiting-time exponent is asserted. In particular the separate conjectural $1/t$ rate in [3] is not obtained here.

There are two difficulties in proving (1.4). First, the ordinary Fisher information need not be bounded uniformly by mass, energy and entropy. We instead estimate $\int\langle v\rangle^{-1} f|\nabla\log f|^2$. Its dissipation controls a weighted Hessian term strongly enough for a quadratic scalar inequality, using the second moment rather than a higher one. The doubled-variable rotations and the even spherical inequality of [5] remain useful, but the one-body weight introduces signed commutators. A near/far decomposition controls them without losing the Hessian damping. This yields positive-time weighted L^3 and unweighted $L^{9/5}$ bounds.

Second, weighted concentration control alone does not rule out a peak moving to infinity. Near $v_0 = Re_1$, a bounded entropy core supplies transverse diffusion of size R^{-1}. We keep those two directions and the full nonlinear collision form. An explicit tangential factorization of its drift gives a Nash inequality involving only transverse Fisher information and collision gradient energy. Combining it with the truncated entropy calculation of [4] and spatial hole filling gives a local small-entropy criterion. Under the exact velocity change $v = v_0 + R^{-1/2}w$, the core diffusion becomes fixed, while the rescaled local positive entropy tends to zero: its mass contributes R^{-2} and its volume factor only $R^{3/2}$. This excludes high-velocity peaks. The remaining bounded region is controlled by a divergence-form Moser estimate using the positive-time global L^2 and L^3 bounds.

The weighted estimate is proved in Section 2, the local criterion in Section 3, and the global bound in Section 4. We write $\langle v\rangle = (1+|v|^2)^{1/2}$, $\kappa_0 = 1/(8\pi)$, and use Lebesgue measure and the convention $0\log 0 = 0$. Norms without a domain are on $\mathbb{R}^3$; matrix inequalities refer to quadratic forms. Constants denoted by C may change between lines, with dependencies stated locally; $\asymp$ means two-sided comparison by such constants. The kernel identities $\operatorname{div}(\Pi/r) = -2z/r^3$ and $\operatorname{div}\operatorname{div}(\Pi/r) = -8\pi\delta_0$ follow by differentiation away from zero and the small-sphere flux. Thus $\operatorname{div} A = \nabla a$ and $\operatorname{div}\operatorname{div} A = -f$, giving the nondivergence equation used above.

2 A weighted Fisher information estimate

The ordinary Fisher information may be large in an entropy-bounded class. The weight $\langle v\rangle^{-1}$ permits a quadratic coercivity estimate using only the second moment. We prove the estimate

first for positive solutions whose logarithmic derivatives have suitable Gaussian integrability, then identify an approximation limit with the given solution.

Put $W(v) = \langle v\rangle^{-1}$ and, on the positive set of a density, write $S = \nabla \log f$ and $H_f = D^2 \log f$. Define

$$J(f) = \int W f |S|^2 \,\mathrm{d}v, \qquad Z(f) = \int \langle v\rangle^{-4} f |H_f|^2 \,\mathrm{d}v. \tag{2.1}$$

At a zero of a smooth nonnegative density, the Fisher integrand is set to zero.

Proposition 2.1. *Let f be a solution as in Theorem 1.1. There is a constant depending only on H such that, for $0 < t \le T$,*

$$J(f(t)) \le C_H(1 + t^{-1}), \qquad \|Wf(t)\|_3 \le C_H(1 + t^{-1}), \qquad \|f(t)\|_{9/5} \le C_H(1 + t^{-2/3}). \tag{2.2}$$

2.1 The lifted first variation

For now $f > 0$. All calculations in this subsection and the next two are made in the following class: f has a Gaussian upper bound, and its logarithmic derivatives grow more slowly than any fixed Gaussian. Such bounds on compact positive-time intervals justify the integrations by parts. We return to the original nonnegative class at the end of this section.

For $z = v - w$ and $r = |z|$, set

$$b_j = e_j \times z, \quad B_j = (b_j, -b_j), \quad a_*(r) = r^{-3/2}, \quad V_j = a_* B_j, \quad Q_6 = \sum_{j=1}^{3} V_j^2.$$

We use the doubled-variable rotations of [5, Sections 3–4]. These are vector fields, acting as directional derivatives on $\mathbb{R}^6$. They are divergence-free and preserve the center $(v + w)/2$ and the relative radius r. For

$$F(v, w) = f(v)f(w), \quad u = \log F, \quad s = \nabla_{v,w} u, \quad M = \mathrm{diag}(W(v)I_3, W(w)I_3),$$

write $\mathcal{I}_M(F) = \iint F s^T M s \,\mathrm{d}v\,\mathrm{d}w$. The identity $\sum_j b_j \otimes b_j = r^2 \Pi(z)$ shows that Q_6 has divergence matrix

$$\begin{pmatrix} K_0 & -K_0 \\ -K_0 & K_0 \end{pmatrix}, \qquad K_0(z) = \Pi(z)/|z|.$$

Consequently $\kappa_0 \int Q_6(f \otimes f)\,\mathrm{d}w = Q(f, f)$, where Q denotes the physical collision operator in (1.1). At the product profile,

$$\frac{\delta \mathcal{I}_M}{\delta F} = \Phi(v) + \Phi(w), \qquad \Phi = -2\,\mathrm{div}(WS) - W|S|^2.$$

The two marginals are equal, and therefore

$$J'[Q(f, f)] = \frac{\kappa_0}{2} \mathcal{I}'_M(F)[Q_6 F]. \tag{2.3}$$

For a divergence-free field V, introduce the matrix expression

$$\mathcal{T}_V M = (DV)M + M(DV)^T - V \cdot \nabla M, \qquad (DV)_{ij} = \partial_j V_i.$$

Differentiating the Fisher integral and integrating once along V gives $\mathcal{I}'_M[VF] = \int F s^T \mathcal{T}_V M s$. Indeed, $\nabla(Vu) = (DV)^T s + Vs$, and the two terms containing Vs cancel. The second density variation is $\mathcal{I}''_M[VF, VF] = 2\int F |\nabla(Vu)|^2_M$. Differentiating the first identity along VF thus yields

$$\mathcal{I}'_M[V^2 F] = -2 \int F |\nabla(Vu)|^2_M + \int F s^T \mathcal{T}_V^2 M s. \tag{2.4}$$

To compute the last term, put $t_* = \nabla_{v,w} a_*$ and $N_j = \mathcal{T}_{B_j} M$. Since $B_j a_* = 0$, $B_j t_* = (DB_j)t_*$, and DB_j is antisymmetric,

$$\mathcal{T}^2_{V_j} M = a_*^2 \mathcal{T}^2_{B_j} M + 2a_*\{B_j \otimes N_j t_* + N_j t_* \otimes B_j\} + 2(t_*^T M t_*) B_j \otimes B_j. \tag{2.5}$$

For completeness, if $J_j x = e_j \times x$, then

$$N_j = \begin{pmatrix} -(b_j \cdot \nabla W(v)) I_3 & (W(v) - W(w)) J_j \\ (W(w) - W(v)) J_j & (b_j \cdot \nabla W(w)) I_3 \end{pmatrix}.$$

One way to check (2.5) is to write $\mathcal{T}_{a_* B_j} M = a_* N_j + B_j \otimes M t_* + M t_* \otimes B_j$. Applying $\mathcal{T}_{a_* B_j}$ again gives the two mixed terms in (2.5); its two remaining rank-one terms give the last positive term.

2.2 Near relative velocities

Let $D_<$ denote the integral in (2.4), summed over j and restricted to $r < \varepsilon$. Write $G_j = a_* B_j u$. Integration of $a_*^2 \mathcal{T}^2_{B_j} M$ along B_j gives

$$\begin{aligned} D_< = & -2 \sum_j \int_{r<\varepsilon} F |\nabla G_j|^2_M + 2 \sum_j \int_{r<\varepsilon} F \frac{t_*^T M t_*}{a_*^2} G_j^2 \\ & + \sum_j \int_{r<\varepsilon} F \{2a_* \nabla G_j \cdot N_j s + 2 G_j t_* \cdot N_j s + a_* G_j s^T N_j s\}. \end{aligned} \tag{2.6}$$

There is no radial boundary flux, since $B_j r = 0$. In this integration, the derivative of F produces the last, cubic-score term. Before adding the mixed term of (2.5), the two terms linear in $N_j s$ are $2a_* \nabla G_j \cdot N_j s - 2 G_j t_* \cdot N_j s$; that mixed term adds $4 G_j t_* \cdot N_j s$. This explains the sign in (2.6).

The spherical estimate used here is [5, Lemma 9.1]: for every positive even C^2 function h on S^2, with $L_j = (e_j \times \sigma) \cdot \nabla_\sigma$,

$$\sum_{i,j} \int_{S^2} h (L_i L_j \log h)^2 \, \mathrm{d}\sigma \geq \frac{19}{4} \sum_j \int_{S^2} h (L_j \log h)^2 \, \mathrm{d}\sigma. \tag{2.7}$$

With $c = (v+w)/2$, the slice $F(c + r\sigma/2, c - r\sigma/2)$ is even. In these coordinates $B_j = 2L_j$ and

$$|\nabla_{v,w} G|^2 = \tfrac{1}{2} |\nabla_c G|^2 + 2 |\nabla_z G|^2.$$

Set $R_c = \langle c \rangle$, $\omega = R_c^{-1}$, and

$$\mathcal{A}_< = \sum_j \int_{r<\varepsilon} \omega F |\nabla G_j|^2, \qquad \mathcal{R}_< = \sum_j \int_{r<\varepsilon} \omega F G_j^2 / r^2.$$

Equation (2.7) gives $\mathcal{A}_< \geq (19/2) \mathcal{R}_<$. For a fixed sufficiently small absolute ε_0 and $\varepsilon \leq \varepsilon_0$,

$$(1-\delta)\omega I_6 \leq M \leq (1+\delta)\omega I_6, \quad \delta \leq C\varepsilon, \quad |N_j| \leq C r \omega / R_c, \quad |t_*|^2 / a_*^2 = 9/(2r^2).$$

The first line of (2.6) is at most

$$-\{2(1-\delta) - (18/19)(1+\delta)\} \mathcal{A}_<.$$

A fixed negative fraction remains for the three commutators. Young's inequality applied to the first two leaves a multiple of

$$P = \iint_{r<\varepsilon} F \frac{\omega}{r R_c^2} |s|^2.$$

For the cubic term, for any small fixed $\eta > 0$,

$$a_* |G_j| |s^T N_j s| \le \eta \omega G_j^2 / r^2 + C_\eta \omega r |s|^4 / R_c^2.$$

Absorbing its first term by the spherical inequality gives

$$D_< \le -c\mathcal{A}_< + C(P + \mathcal{Q}), \qquad \mathcal{Q} = \iint_{r<\varepsilon} F \frac{\omega r}{R_c^2} |s|^4. \tag{2.8}$$

Apply the Sobolev inequality to $\sqrt{Wf}$ to get $B := \|Wf\|_3 \le C(1 + J)$. On $r < 1$ the three weights at v, w, c are comparable. Splitting the local Coulomb potential at a radius ρ gives

$$\int_{|w-v|<\varepsilon} \frac{f(w)}{|v-w|} \, \mathrm{d}w \le C\langle v\rangle B\rho + \rho^{-1} \le C(\langle v\rangle B)^{1/2}$$

when the optimizing radius is at most ε. If it is larger, the near estimate on the entire ball gives the same last bound. Therefore

$$P \le CB^{1/2} \int f |S|^2 \langle v\rangle^{-5/2} \, \mathrm{d}v \le C(1 + J^{3/2}).$$

Mass and energy give $\int_{|w-v|<1} f(w) \, \mathrm{d}w \le C\langle v\rangle^{-2}$. Hence $\mathcal{Q} \le C\varepsilon \int \langle v\rangle^{-5} f |S|^4 \, \mathrm{d}v$. Recall that $S = \nabla \log f$, $H_f = D^2 \log f$ and $Z = \int \langle v\rangle^{-4} f |H_f|^2$. For $\beta(v) = \langle v\rangle^{-5}$, integration of $\operatorname{div}(\beta f |S|^2 S)$ gives

$$\int \beta f |S|^4 = -\int \beta f \{|S|^2 \operatorname{tr} H_f + 2S^T H_f S + (\nabla \log \beta \cdot S)|S|^2\}.$$

Using $|\nabla \log \beta| \le 5/\langle v\rangle$ and Young's inequality shows $\int \beta f |S|^4 \le CZ + C$. We conclude that

$$D_< \le -c\mathcal{A}_< + C(1 + J^{3/2}) + C\varepsilon Z. \tag{2.9}$$

The constants here are independent of decreasing $\varepsilon \le \varepsilon_0$.

2.3 Separated coercivity and a scalar inequality

Mass, energy and entropy supply a good set inside a fixed ball. To see this explicitly, energy puts at least three quarters of the mass in B_{R_0}. The negative entropy is bounded by mass and energy, since $-x \log x \le \langle v\rangle x + e^{-\langle v\rangle - 1}$ for $0 \le x \le 1$. Thus the positive entropy is bounded by H plus an absolute constant. Choose ℓ small and N large so the portions $f < \ell$ inside this ball and $f > N$ each have mass at most $1/8$. The remaining set has mass at least $1/2$, and hence volume at least $1/(2N)$. In particular

$$E = \{w \in B_{R_0} : f(w) \ge \ell\} \quad \text{has volume at least } \mu_0 > 0,$$

with R_0, ℓ, μ_0 depending only on H.

Choose $d > 0$ so that every radius-d tube around a line has volume at most $\mu_0/2$ in B_{R_0}. For each unit e and each v, a subset of E of volume at least $\mu_0/2$ is distance at least d from $v + \mathbb{R}e$. On this set

$$|v-w|^{-1} e^T \Pi(v-w) e = \frac{\operatorname{dist}(w, v + \mathbb{R}e)^2}{|v-w|^3} \ge c_H \langle v\rangle^{-3}.$$

Consequently, for every $0 < \varepsilon \le d/2$,

$$\int_{|v-w|\ge\varepsilon} \frac{\Pi(v-w)}{|v-w|} f(w)\,\mathrm{d}w \ge c_H \langle v\rangle^{-3} I_3, \tag{2.10}$$

with a lower constant independent of ε.

Return to $G_j = r^{-3/2} B_j \log(f(v)f(w))$ and $M = \operatorname{diag}(W(v)I_3, W(w)I_3)$. In the v-gradient of G_j, the Hessian term is $a_* H_f(v) b_j$ and the remainder is bounded by $Cr^{-3/2}|S(v) - S(w)|$. Using (2.10) and $W(v)/W(w) \le 1 + r$ gives

$$\sum_j \iint_{r\ge\varepsilon} F |\nabla G_j|_M^2 \ge c_H Z - C_\varepsilon J.$$

Conjugation by $M^{-1/2}$ bounds the three terms of (2.5) by

$$Cr^{-3}(1+r)^2, \qquad Cr^{-3}(1+r)^{3/2}, \qquad Cr^{-3}(1+r),$$

respectively. They are bounded on $r \ge \varepsilon$, so the separated contribution obeys

$$D_{\ge} \le -c_H Z + C_\varepsilon J. \tag{2.11}$$

The lower constant is independent of decreasing ε; the positive constant need not be.

First fix the entropy good set and d, then choose $\varepsilon \le d/2$ sufficiently small to absorb the last term of (2.9) into (2.11). Combining with (2.3),

$$J' \le -c_H Z + C_H(1 + J^{3/2}). \tag{2.12}$$

On the other hand, integration by parts gives

$$J = -\int W f \operatorname{tr} H_f - \int f \nabla W \cdot S \le C Z^{1/2} + C J^{1/2},$$

since $\int f W^2 \langle v\rangle^4 = \int f \langle v\rangle^2 = 4$. Thus $Z \ge cJ^2 - C$. Above a threshold depending only on H, (2.12) implies $J' \le -a_H J^2/2$. Integrating the reciprocal on each above-threshold interval yields

$$J(f(t)) \le C_H(1 + (t-s)^{-1}), \qquad 0 < s < t. \tag{2.13}$$

Letting $s \downarrow 0$ proves the first estimate in (2.2) in the current integration class.

The singular integrations can be performed first on $\delta < r < R$ with bounded center. The rotations preserve both boundaries. Near $r = 0$, $S(v) - S(w) = O(r)$, so $G_j = O(r^{1/2})$ and $\nabla G_j = O(r^{-1/2})$, giving integrable relative kernels. At infinity the Gaussian bounds dominate all the displayed terms. These observations justify removal of the cutoffs.

2.4 The original nonnegative solution class

Completion of the proof of Proposition 2.1. Fix one solution f of the original problem and let $f_0 = f(0)$. In the following approximation argument, auxiliary constants may depend on this individual datum. They will be absent from the estimate transferred to f.

First $i(f_0) = \int |\nabla f_0|^2/f_0$ is finite, even if f_0 vanishes. For any $N > 3$, the nonnegative function $h = \langle v\rangle^N f_0$ has bounded Hessian. Taylor's formula along a unit direction gives $|\partial_e h|^2 \le 2\|D^2 h\|_\infty h$;

choose the step $-\partial_e h/\|D^2 h\|_\infty$, with the zero-Hessian case treated directly. Expanding the derivative of $\langle v\rangle^{-N} h$ yields

$$\frac{|\nabla f_0|^2}{f_0} \le C_{f_0,N}\langle v\rangle^{-N} + C_N f_0 \langle v\rangle^{-2},$$

which is integrable. At zeros of f_0 its gradient vanishes.

Choose a smooth χ_R equal to one on B_R, supported in B_{2R}, with gradient bounded by C/R. For a small fixed $\beta > 0$, form

$$g_R = \chi_R^2 f_0 + \epsilon_R \phi_\beta, \quad \phi_\beta = (\beta/\pi)^{3/2} e^{-\beta|v|^2}, \quad h_R = g_R / \int g_R.$$

Let $m_R = \int v h_R$ and $\theta_R = \frac{1}{3}\int |v - m_R|^2 h_R$. Set

$$f_{0,R}(v) = \theta_R^{3/2} h_R(m_R + \sqrt{\theta_R} v). \tag{2.14}$$

For large R these are positive smooth profiles with mass one, mean zero and energy three; $m_R \to 0$, $\theta_R \to 1$. Outside a compact set they agree with a shifted isotropic Gaussian.

The pointwise Cauchy–Schwarz inequality gives $i(g+h) \le i(g) + i(h)$ and

$$i(\chi_R^2 f_0) \le 2i(f_0) + 8\int f_0 |\nabla \chi_R|^2.$$

For each chosen β, take $\epsilon_R \downarrow 0$ so fast that its contributions to Fisher information, the initial supremum and any fixed moments needed below are bounded by one and tend to zero. Thus the normalized family has uniformly bounded ordinary Fisher information and every such moment, in terms of f_0 alone. Write $M_k(h) = \int \langle v\rangle^k h(v)\,\mathrm{d}v$. Choose a moment exponent k_* sufficiently large for [5, Theorem 2.3], and also $k_* \ge 18$; both M_{18} and M_{k_*} are controlled. The bounds can be fixed before choosing β small enough for Gaussian propagation, since decreasing the added Gaussian mass compensates for its broadening.

Let $\mathcal{P}_2$ be the probability measures with finite second moment and let $W_2(\mu,\nu)^2$ be the infimum of $\int |v-w|^2\,\mathrm{d}\pi(v,w)$ over their probability couplings π. The profiles converge to f_0 in W_2. Indeed, $\int (1+|v|^2)|h_R - f_0|\,\mathrm{d}v \to 0$. Couple the common part identically and couple the two remaining equal-mass portions by their normalized product. The quadratic cost tends to zero by this weighted L^1 convergence. Applying the affine map in (2.14) adds a cost tending to zero, since $m_R \to 0$, $\theta_R \to 1$ and the second moments stay bounded.

Also $\int f_{0,R}\log f_{0,R} \to \int f_0 \log f_0$. On compact sets this follows by dominated convergence and the uniform individual-datum supremum bound. Positive entropy tails are bounded by a constant times the mass tails. For the negative tails use $-x\log x \le \langle v\rangle x + e^{-\langle v\rangle - 1}$; outside B_L the first term is $O(L^{-1})$ by energy. Mass normalization and the affine map change entropy only by terms tending to zero. In particular the approximants have entropy at most $H+1$ for large R.

Guillen–Silvestre's global existence theorem [5, Theorem 1.2] supplies the actual nonlinear solution f_R from each $f_{0,R}$. Their source equation is the physical equation with source time $s = \kappa_0 t$. Their ordinary Fisher monotonicity and moment propagation [5, Theorems 1.1 and 2.4] give, on any fixed finite interval, uniform bounds for $i(f_R)$, $\|f_R\|_3$, $M_{18}(f_R)$ and $M_{k_*}(f_R)$ depending on f_0 and the interval. These bounds are used for identification only.

The collision matrices are bounded above by splitting the potential into a unit ball and its complement and using the auxiliary L^3 bound. The entropy good set gives $A[f_R] \ge c_H \langle v\rangle^{-3} I_3$ and hence

$$\det A[f_R] \ge \delta(1+|v|)^{-9}, \qquad \int (1+|v|)^9 f_R^2\,\mathrm{d}v \le C\Big(\int f_R^3\Big)^{1/2}\Big(\int \langle v\rangle^{18} f_R\Big)^{1/2} \le N_*.$$

Moreover $f_{R,t} = A[f_R] : D^2 f_R + f_R^2$. Apply [7, Theorem 2.1] with

$$d = 3, \quad p = 2, \quad \alpha = 1 < 4/3, \quad \beta = 0, \quad \kappa = 9.$$

Its hypotheses are the just established determinant lower bound, matrix upper bound and weighted L^2 bound, with $\beta \geq -\kappa/d$. Each approximation is individually bounded and classical on every unit interval, as required by that theorem. It gives a fixed short interval of length $h > 0$, depending on the auxiliary bounds, on which the bound is $Ct^{-3/4}$. Apply it from zero for $t \leq h$ and from $t - h$ for $t > h$, using the global approximants and their auxiliary bounds on $[0, T + 1]$. We obtain

$$\|f_R(t)\|_\infty \leq C_{f_0,T}(1 + t^{-3/4}), \qquad 0 < t \leq T. \tag{2.15}$$

In particular the time integrals of these suprema are uniformly bounded.

We also need the logarithmic integrations leading to (2.13). The initial suprema are bounded uniformly in R. For $M(t) = \|f_R(t)\|_\infty$, the maximum principle gives the upper Dini inequality $M' \leq M^2$, which bounds them on a common short interval from zero. Combining with (2.15) gives a uniform auxiliary supremum on $[0, T]$. Together with M_{k_*} this supplies the hypotheses of [5, Theorem 2.3]. Choose the fixed initial β sufficiently small, leaving room for $\theta_R \to 1$. In the translated tail coordinates the initial upper and lower Gaussians have the same exponent; that theorem propagates the upper Gaussian, and [5, Propositions B.1–B.2] give a lower exponent arbitrarily close to it and corresponding derivative bounds. Their following logarithmic-derivative statement justifies all preceding computations for each f_R on compact positive-time intervals. The Gaussian prefactors can depend on R and are not used in (2.13).

For the identification we use [2, Theorem 2(ii)]. Its weak solutions are probability curves in $L^\infty_t \mathcal{P}_2 \cap L^1_t L^\infty_v$. If initial data converge in W_2 and the approximants' $L^1_t L^\infty_v$ norms are uniformly bounded, then the solutions converge uniformly in time in W_2. The given classical solution has an individual finite $L^1_t L^\infty_v$ norm on its compact classical interval. With $K_0(z) = \Pi(z)/|z|$, the source generator is

$$\tfrac{1}{2} K_0(v - w) : D^2\varphi(v) - 2(v - w)|v - w|^{-3} \cdot \nabla\varphi(v).$$

Multiplication by $2\kappa_0$ gives the physical weak generator, so the source time here is $s = 2\kappa_0 t$. All its hypotheses are satisfied by (2.14) and (2.15). Consequently

$$\sup_{0 \leq t \leq T} W_2(f_R(t), f(t)) \longrightarrow 0. \tag{2.16}$$

The weighted Fisher functional has the dual formula

$$J(h) = \sup_{\xi \in C_c^\infty(\mathbb{R}^3;\mathbb{R}^3)} \int h\{-2 \operatorname{div}(W\xi) - W|\xi|^2\} \, \mathrm{d}v. \tag{2.17}$$

On the positive set write $S_h = \nabla \log h$. Integration by parts and $2S_h \cdot \xi - |\xi|^2 \leq |S_h|^2$ give one inequality. For the reverse, use compact cutoffs times $\nabla h/(h + \epsilon)$ and remove the cutoff and ϵ monotonically. The coefficient of $Wh|S_h|^2$ tends to one and lies between zero and one. This proves the formula also with the extended-energy convention. Each test integrand is compact smooth and globally Lipschitz; the coupling inequality gives convergence of its integral under (2.16). Taking the supremum in (2.17),

$$J(f(t)) \leq \liminf_R J(f_R(t)) \leq C_{H+1}(1 + t^{-1}).$$

The auxiliary Fisher, moment, supremum and Gaussian constants have disappeared. Rename the permitted function C_{H+1} as C_H.

Finally $\sqrt{Wf} \in H^1$: regularize it by $\sqrt{Wf+\epsilon}-\sqrt{\epsilon}$, whose function and gradient converge in L^2 by the finite J and mass. Sobolev gives $\|Wf\|_3 \le C(J+1)$. The exact factorization

$$f^{9/5} = (f^3\langle v\rangle^{-3})^{2/5}(f\langle v\rangle^2)^{3/5}$$

and Hölder, together with $\int f\langle v\rangle^2 = 4$, give $\|f\|_{9/5} \le C\|Wf\|_3^{2/3}$. This proves (2.2) for the entire original class. □

3 A local regularity lemma

Near a large velocity, a bounded core provides two useful diffusion directions. The following lemma retains those directions together with the nonlinear collision energy. Its constants depend on the potentials divided by the cutoff scales, which allows the local ball to grow after velocity dilation.

Write $v = (x, y) \in \mathbb{R} \times \mathbb{R}^2$ and $P_y = \operatorname{diag}(0, 1, 1)$. For a density F, let

$$P_F(v) = \int \frac{F(w)}{|v-w|}\,\mathrm{d}w, \qquad c_{2,F}(v) = \int \frac{F(w)\wedge 2}{|v-w|^2}\,\mathrm{d}w.$$

Proposition 3.1. *Let F be a nonnegative smooth rapidly decaying solution of (1.1) on $[a,b]\times\mathbb{R}^3$, and let $L \ge 1$. Suppose that, at every time, a measurable set $C \subset \mathbb{R}^3 \setminus B_L$ satisfies*

$$A[F\mathbf{1}_C](v) \ge cP_y \quad (v \in B_L), \qquad c > 0. \tag{3.1}$$

Assume also

$$\sup_{[a,b]\times B_L} P_F \le \Lambda, \qquad \sup_{[a,b]\times B_L} c_{2,F} \le \Xi, \qquad \Lambda/L^2 + \Xi/L \le K_*. \tag{3.2}$$

For each $0 < \theta < b-a$ there exists $\epsilon_*(c, K_*, b-a, \theta) > 0$ such that

$$\int_a^b \int_{B_L} F(\log_+ F + \log_+^2 F)\,\mathrm{d}v\,\mathrm{d}t \le \epsilon_* \quad \Longrightarrow \quad F \le 2 \quad \text{on } [a+\theta, b]\times B_{L/2}. \tag{3.3}$$

No normalization of the total mass of F is imposed.

3.1 A tangential factorization and a Nash inequality

The collision drift can be expressed using only the two y-derivatives of a density. For $z = (x, y)$, $r = |z|$, $a_x = |x|$ and $\sigma = \operatorname{sgn} x$, define, for $x \ne 0$,

$$\mathcal{K}(z) = \begin{pmatrix} \sigma y^T/[r(r+a_x)] \\ -I_2/(r+a_x) + yy^T/[r(r+a_x)^2] \end{pmatrix}. \tag{3.4}$$

Its values on $x = 0$ are immaterial for integration. Direct multiplication and differentiation give

$$z^T\mathcal{K} = 0, \quad \mathcal{K}^T\mathcal{K} = I_2/(r+a_x)^2, \quad \mathcal{K}\mathcal{K}^T = \Pi(z)/(r+a_x)^2, \quad \operatorname{div}_y \mathcal{K} = z/r^3. \tag{3.5}$$

To check the first three, let $e_r = y/|y|$ and let e_θ be perpendicular to it in $\mathbb{R}^2$. Their images are $(\sigma|y|, -a_x e_r)/[r(r+a_x)]$ and $(0, -e_\theta)/(r+a_x)$, two orthogonal tangent vectors of equal length. The first-row divergence is x/r^3. For the lower block, differentiation of $qI_2 + dyy^T$, with $q = -1/(r+a_x)$ and $d = 1/[r(r+a_x)^2]$, gives y/r^3, using $|y|^2 = (r-a_x)(r+a_x)$.

The last identity is distributional on $\mathbb{R}^3$. A jump across $x = 0$ contributes no y-divergence delta. At zero $\mathcal{K} = O(r^{-1})$, so the small-sphere flux is $O(\epsilon)$; the ordinary derivative is $O(r^{-2})$ and is locally integrable. Therefore, with $A_0[h] = (\Pi/r) * h$ and $b_0[h] = \operatorname{div} A_0[h]$,

$$b_0[h] = -2\mathcal{K} * \nabla_y h.$$

For every vector ξ, Cauchy–Schwarz and $r/(r + a_x)^2 \le 1/r$ imply

$$|\xi \cdot b_0[h](v)|^2 \le 4\xi^T A_0[h](v)\xi \; p(v), \qquad p(v) = \int \frac{|\nabla_y h(w)|^2/h(w)}{|v - w|}\,\mathrm{d}w. \tag{3.6}$$

No inverse matrix is needed in this estimate.

Let $M_h = \int h$ and $I_y(h) = \int |\nabla_y h|^2/h$. The marginal $\rho(y) = \int h(x, y)\,\mathrm{d}x$ satisfies $I(\rho) \le I_y(h)$. In two dimensions, $\operatorname{div}_y((y - y_0)/|y - y_0|) = 1/|y - y_0|$, without a delta at zero. Integration by parts gives

$$\sup_v P_h(v) \le \sup_{y_0} \int \frac{\rho(y)}{|y - y_0|}\,\mathrm{d}y \le \int |\nabla\rho|\,\mathrm{d}y \le \sqrt{M_h I_y(h)}. \tag{3.7}$$

For compact bounded profiles the boundary passages follow directly; for the current decaying profiles the infinity cutoff error tends to zero by integrability. In particular Tonelli gives $\int hp \le M_h^{1/2} I_y(h)^{3/2}$.

Lemma 3.2. *Let h be a nonnegative density. With $E[h] = \int (\nabla h)^T A[h]\nabla h/h$, one has*

$$\left(\int h^2\,\mathrm{d}v\right)^2 \le \frac{1}{2\pi} M_h^{1/2} I_y(h)^{3/2} E[h]. \tag{3.8}$$

It holds for the smooth rapidly decaying profiles and the compact square-root level truncations used below, whenever the displayed energies are finite.

Proof. Since $\operatorname{div} b_0[h] = -8\pi h$,

$$8\pi \int h^2 = \int \nabla h \cdot b_0[h] \le 2\{E[h]/\kappa_0\}^{1/2} \left(\int hp\right)^{1/2}.$$

Equations (3.6)–(3.7) and $\kappa_0 = 1/(8\pi)$ give (3.8). Zero mass is trivial. If $\sqrt{h}$ is compactly supported Lipschitz, use $\nabla h = 2\sqrt{h}\nabla\sqrt{h}$ in these same integrations. The kernels are locally integrable and their potentials are bounded on the support, so weak integration by parts or smooth approximation gives the identical formulas. Below $\sqrt{h} = \chi(\sqrt{F} - \sqrt{k})_+$ with $k \ge 1$, which is of this class. □

3.2 The local collision energy and entropy

For $1 \le k \le 2$ put

$$g = (F - k)_+, \quad u = (\sqrt{F} - \sqrt{k})_+, \quad q = u^2, \quad V = \mathbf{1}_{\{F>k\}} \nabla \log F,$$

$$h_k(F) = (F \log(F/k) - F + k)\mathbf{1}_{\{F>k\}}, \qquad W_k(F) = F\{\log_+(F/k) + \log_+^2(F/k)\}.$$

Then $q \le h_k \le W_k$ and $g \le F \log_+(F/k)$. For $0 \le \psi \le 1$ smooth and supported in B_L, set

$$D_\psi = \frac{1}{2} \iint F F_* (\psi V - \psi_* V_*)^T K(v - w)(\psi V - \psi_* V_*),$$

$$E_\psi = \int \psi^2 F V^T A[F] V, \qquad C_\psi = \iint \psi(v)\nabla g(v)^T K(v - w)\psi(w)\nabla g(w),$$

where a star means evaluation at w and $K = \kappa_0 \Pi / r$ is the physical kernel. Symmetry gives $D_\psi = E_\psi - C_\psi \geq 0$. Restrict the positive pair integral to $B_L \times C$ and $C \times B_L$. As $\psi = 0$ on C, (3.1) gives

$$D_\psi \geq \int \psi^2 F V^T A[F \mathbf{1}_C] V \geq c \int \psi^2 \mathbf{1}_{\{F>k\}} \frac{|\nabla_y F|^2}{F}. \tag{3.9}$$

For a cutoff with $|\nabla \psi| \leq C\delta^{-1}$ and $|D^2 \psi| \leq C\delta^{-2}$, put $G = \psi g$. Expanding $\psi \nabla g = \nabla G - g \nabla \psi$ in both variables gives

$$\begin{aligned} C_\psi = & \iint \nabla G_v^T K \nabla G_w - \iint \nabla G_v^T K g_w \nabla \psi_w \\ & - \iint g_v \nabla \psi_v^T K \nabla G_w + \iint g_v g_w \nabla \psi_v^T K \nabla \psi_w. \end{aligned}$$

Here subscripts indicate the evaluation variable. With $b_K = \operatorname{div} K$, the first term is

$$- \iint G_v G_w \operatorname{div} \operatorname{div} K = \int G^2.$$

Integrating the two middle terms in v and w, respectively, gives the sum

$$\iint G_v g_w b_K \cdot \nabla \psi_w - \iint g_v G_w b_K \cdot \nabla \psi_v.$$

Thus

$$\begin{aligned} C_\psi = \int \psi^2 g^2 + \iint g g_* \{ & \nabla \psi(v)^T K \nabla \psi(w) \\ & + b_K \cdot [\psi(v) \nabla \psi(w) - \psi(w) \nabla \psi(v)] \}. \end{aligned} \tag{3.10}$$

Here $\operatorname{div} \operatorname{div} K = -\delta_0$. The bracket in the b_K term is bounded by $C\delta^{-2}|v - w|$, and both error variables lie in B_L. Symmetrizing $2g(v)g(w) \leq g(v)^2 F(w)/F(v) + g(w)^2 F(v)/F(w)$, with zero values on $F = 0$, shows

$$\iint_{B_L^2} \frac{g g_*}{|v - w|} \, \mathrm{d}v \, \mathrm{d}w \leq \Lambda \int_{B_L} g^2 / F \leq 4\Lambda \int_{B_L} q.$$

Thus

$$E_\psi \leq D_\psi + \int \psi^2 g^2 + C\Lambda \delta^{-2} \int_{B_L} q. \tag{3.11}$$

The linear dependence on the truncation mass will allow a superlinear level recurrence.

We next give the entropy estimate at the cutoff scales used here. The calculation follows [4, (3.29)–(3.34)], before its subsequent isotropic coercivity step. We give it for smooth solutions and arbitrary δ, since the final small-radius weak-solution lemma of that paper has different geometric restrictions.

Lemma 3.3. *For a nondecreasing time cutoff $0 \leq \eta \leq 1$ with $\eta(a) = 0$,*

$$\sup_t \eta(t) \int \psi^2 h_k(F(t)) + c_0 \int_a^b \eta D_\psi \, \mathrm{d}t \leq C(\|\eta'\|_\infty + 1 + \Lambda \delta^{-2} + \Xi \delta^{-1}) \int_{[a,b] \times \operatorname{supp} \psi} W_k(F). \tag{3.12}$$

The constants are absolute, apart from the fixed cutoff shapes.

Proof. Let $l = F \wedge k$, $d_K = -\operatorname{div}\operatorname{div} K = \delta_0$ and $H_\psi = \int \psi^2 h_k(F)$. Testing the divergence equation against $\psi^2 \log_+(F/k)$ splits $H'_\psi = T_1 + T_2$, with the derivatives falling on the logarithm and cutoff, respectively. Before symmetrization,

$$\begin{aligned} T_1 = & -\iint \psi_v^2 V_v^T K(v-w)[F_w \nabla F_v - F_v \nabla F_w] \\ = & -E_\psi + \iint \psi_v^2 \nabla g_v^T K \nabla g_w + \iint \psi_v^2 \nabla g_v^T K \nabla l_w. \end{aligned}$$

We used $FV = \nabla g$ and $\nabla F = \nabla g + \nabla l$. Subtracting C_ψ from the high–high integral and using $E_\psi - C_\psi = D_\psi$ leaves $\frac{1}{2}\iint (\psi_v - \psi_w)^2 \nabla g_v^T K \nabla g_w$ as its remainder. Integrating this remainder and the high–low integral twice gives

$$T_1 = -D_\psi + \iint g g_* \Gamma_1 + \iint g(v) l(w) \Gamma_2,$$

$$\begin{aligned} \Gamma_1 &= \tfrac{1}{2} d_K(v-w)(\psi_v - \psi_w)^2 - b_K \cdot (\nabla\psi_v + \nabla\psi_w)(\psi_v - \psi_w) - \nabla\psi_v^T K \nabla \psi_w, \\ \Gamma_2 &= d_K(v-w)\psi_v^2 - b_K \cdot \nabla(\psi_v^2). \end{aligned}$$

The delta term in Γ_1 vanishes. The other terms are bounded by $C\delta^{-2}/r$, with at least one variable in $\operatorname{supp}\psi$. They contribute at most $C\Lambda\delta^{-2}\int_{\operatorname{supp}\psi} g$. The delta in Γ_2 gives $k\int\psi^2 g$; its remaining term is bounded by $C\Xi\delta^{-1}\int_{\operatorname{supp}\psi} g$, because $\int (F\wedge k)(w)/|v-w|^2\,\mathrm{d}w \le c_{2,F}(v)$.

Write $\ell_v = \log_+(F(v)/k)$. In T_2, the high part is

$$\begin{aligned} T_2^+ = & -2\iint F F_* K(\psi_v V_v - \psi_w V_w)\cdot \ell_v \nabla\psi_v \\ & + 2\int F\ell_v \nabla\psi_v \cdot \int \Gamma_3(v,w) g(w)\,\mathrm{d}w\,\mathrm{d}v, \\ \Gamma_3(v,w) = & \, b_K(v-w)(\psi_v - \psi_w) + K(v-w)\nabla\psi_w. \end{aligned}$$

Cauchy–Schwarz in the pair measure bounds the first term by $D_\psi/2 + C\Lambda\delta^{-2}\int F\ell^2$. Since $|\Gamma_3| \le C\delta^{-1}/r$, the second is bounded by $C\Lambda\delta^{-2}\int F\ell$, with v in $\operatorname{supp}\psi$. The low part is $\int (b_K * l)\cdot F\ell\nabla(\psi^2)$ and is bounded by $C\Xi\delta^{-1}\int F\ell$ on the same support. These estimates and $g \le F\ell$, $k \le 2$ give the differential entropy inequality. Multiply by η and integrate. The term $\eta' H_\psi$ is bounded by $\|\eta'\|_\infty \int W_k$, giving (3.12) after taking a supremum and the final-time energy bound.

The truncations have the stated weak derivatives; the square root is used only above $k \ge 1$. Smooth truncation approximations pass to the limit using local smoothness, bounded level derivatives and the locally integrable kernels r^{-1}, r^{-2}. The only double-divergence distribution is the explicitly evaluated delta. No relation between δ and the time length, and no restriction $\delta \le 1$, has entered this calculation. □

3.3 Spatial hole filling

Fix a time and level k, and recall $u = (\sqrt{F} - \sqrt{k})_+$ and $q = u^2$. For $r < s$, choose $\chi = 1$ on B_r, supported in an intermediate ball, and $\psi = 1$ on $\operatorname{supp}\chi$, supported in B_s. Their derivative length δ is comparable to $s - r$. Put

$$M_q = \int_{B_s} q, \quad I_q = 4\int_{B_s} |\nabla_y u|^2, \quad J_\delta = C(I_q + \delta^{-2} M_q), \quad K_\delta = k + \Lambda\delta^{-2}, \qquad Z(r) = \int_{B_r} q^2.$$

Apply Lemma 3.2 to $h = \chi^2 q$. Its mass is at most M_q and its transverse information is at most J_δ. Matrix positivity gives $A[h] \le A[F]$. Since $\nabla u = \mathbf{1}_{\{F>k\}} \nabla F/(2\sqrt{F})$, product differentiation gives

$$E[h] \le C(E_\psi + \Lambda \delta^{-2} M_q).$$

The first term bounds the $\chi \nabla u$ contribution, while the second bounds $u \nabla \chi$. Using (3.11) and $g^2 \le 2q^2 + 8kq$,

$$Z(r)^2 \le C M_q^{1/2} J_\delta^{3/2} [D_\psi + Z(s) + K_\delta M_q].$$

Take square roots and use Young on the term containing $Z(s)$. For any fixed $0 < \epsilon < 1$,

$$Z(r) \le \epsilon Z(s) + C_\epsilon [M_q^{1/2} J_\delta^{3/2} + M_q^{1/4} J_\delta^{3/4} D_\psi^{1/2} + K_\delta^{1/2} M_q^{3/4} J_\delta^{3/4}]. \tag{3.13}$$

Choose $\rho_j = R_o - (R_o - R_i)2^{-j}$ between fixed interior radii $R_i < R_o$. Iterate (3.13), using common mass and transverse information on B_{R_o} and gaps $\delta_j \asymp (R_o - R_i)2^{-j}$. The residual $\epsilon^N Z(\rho_N)$ tends to zero for each individual smooth profile, since it is bounded on B_{R_o}. Interpolation $\int_{B_{R_i}} q^{5/3} \le M_q^{1/3} Z(R_i)^{2/3}$ and concavity give

$$\int_{B_{R_i}} q^{5/3} \le C_\epsilon \sum_{j \ge 0} \epsilon^{2j/3} [M_q^{2/3} J_j + M_q^{1/2} J_j^{1/2} D_j^{1/3} + K_j^{1/3} M_q^{5/6} J_j^{1/2}], \tag{3.14}$$

where $J_j = C(I_q + \delta_j^{-2} M_q)$, $K_j = k + \Lambda \delta_j^{-2}$ and D_j is the actual energy of the jth cutoff. All terms are nonnegative, so this series may now be integrated in time.

3.4 The level iteration

Proof of Proposition 3.1. Set

$$k_n = 2 - 2^{-n}, \quad r_n = L(1+2^{-n})/2, \quad t_n = a + \theta(1-2^{-n}), \quad \mathcal{J}_n = (t_n, b), \quad Z_n = \int_{\mathcal{J}_n \times B_{r_n}} W_{k_n}(F).$$

Then $Z_1 \le Z_0$. For $n \ge 1$, take a time cutoff vanishing before t_{n-1} and equal to one after t_n, with derivative bounded by $C2^n/\theta$. Apply Lemma 3.3 on $[t_{n-1}, b]$. A base spatial cutoff equals one on B_{r_n} and is supported in $B_{r_{n-1}}$. Since $W_{k_n} \le W_{k_{n-1}}$, (3.9) gives, for $A_n = C_0 4^n Z_{n-1}$,

$$\sup_{\mathcal{J}_n} M_q \le A_n, \qquad \int_{\mathcal{J}_n} I_q \, \mathrm{d}t \le A_n, \quad M_q(t) = \int_{B_{r_n}} q_{k_n}, \quad I_q(t) = 4 \int_{B_{r_n}} |\nabla_y u_{k_n}|^2. \tag{3.15}$$

Here and below the constants depend only on $c, K_*, b-a, \theta$.

Use (3.14) between r_{n+1} and r_n; its gaps are $\delta_{n,j} \asymp L 2^{-n-j}$. Applying (3.12) to each hole cutoff on the same time interval gives

$$\int_{\mathcal{J}_n} D_j \, \mathrm{d}t \le C 4^j A_n, \qquad \int_{\mathcal{J}_n} J_j \, \mathrm{d}t \le C_{b-a} 4^{n+j} A_n, \qquad K_j \le C 4^{n+j}. \tag{3.16}$$

Indeed $\Lambda \delta_{n,j}^{-2} \le C K_* 4^{n+j}$ and $\Xi \delta_{n,j}^{-1} \le C K_* 2^{n+j}$, while $L \ge 1$. The three terms in the series (3.14) have time integrals bounded by

$$C_{b-a} 4^{n+j} A_n^{5/3}, \qquad C_{b-a} 4^{n/2+5j/6} A_n^{4/3}, \qquad C_{b-a} 4^{5(n+j)/6} A_n^{4/3},$$

respectively. These follow from (3.15)–(3.16), using Hölder exponents $2, 3, 6$ for the middle term and Cauchy–Schwarz for the last. Their growth in j is at most 4^j. Fix $\epsilon = 1/256$ in (3.13); then $4\epsilon^{2/3} < 1$. Summation yields a fixed $B > 1$ such that

$$\int_{\mathcal{J}_n \times B_{r_{n+1}}} q_{k_n}^{5/3} \le CB^n (Z_{n-1}^{5/3} + Z_{n-1}^{4/3}).$$

Each cutoff energy was integrated separately in (3.16).

On $F > k_{n+1}$ the level gap gives $F \le C4^n q_{k_n}$. Since $F(\log_+ F + \log_+^2 F) \le CF^{5/3}$ for $F \ge 1$, we obtain, after enlarging B,

$$Z_{n+1} \le CB^n (Z_{n-1}^{4/3} + Z_{n-1}^{5/3}), \qquad n \ge 1. \tag{3.17}$$

One may take $B = 4^6$: before the level comparison the growth is at most $4^{8n/3}$, and that comparison adds at most $C4^{5n/3}$. Choose $0 < d \le 1$ so that $2Cd^{1/3}B^7 \le 1$, and require $Z_0 \le dB^{-3}$. The two base cases follow from $Z_1 \le Z_0$; substitution in (3.17) gives $Z_n \le dB^{-3n}$. Fatou on $(a+\theta, b) \times B_{L/2}$ gives $W_2(F) = 0$, hence $F \le 2$. Continuity includes the time endpoints. This proves the proposition with constants independent of L, Λ, Ξ separately. □

4 A global density bound

We now verify the hypotheses of Proposition 3.1 near every sufficiently distant velocity. The local mass decreases like R^{-2}, whereas the volume factor in the useful velocity dilation grows only like $R^{3/2}$. This leaves the small entropy needed for the level iteration.

Fix $0 < \tau < T$. By Proposition 2.1, uniformly for $\tau/4 \le t \le T$,

$$\|f(t)/\langle v\rangle\|_3 + \|f(t)\|_{9/5} \le B(H, \tau). \tag{4.1}$$

For a fixed $d > 0$, energy gives

$$\int_{B_d(v_0)} f(t, v)\, \mathrm{d}v \le C_d R^{-2}, \qquad |v_0| = R \ge 2d + 2. \tag{4.2}$$

These two estimates also control positive log-squared entropy on that ball:

$$\int_{B_d(v_0)} f(\log_+ f + \log_+^2 f)\, \mathrm{d}v \le C_{H,\tau,d} R^{-2} \log^2 R. \tag{4.3}$$

Indeed let m be the mass of $f\mathbf{1}_{B_d(v_0)\cap\{f>1\}}$. If $m = 0$ there is nothing to prove. Under the probability measure $m^{-1} f\, \mathrm{d}v$ on this set, $Z = \log f$ satisfies $\mathbb{E}e^{(4/5)Z} \le C_B/m$ by (4.1). Thus $\mathbb{P}(Z > s) \le \min(1, C_B m^{-1} e^{-4s/5})$. Integrating $2s\mathbb{P}(Z > s)$ gives $\mathbb{E}Z^2 \le C[1 + \log^2(C_B/m)]$. The first logarithm has the same bound. For small m, the function $m[1 + \log^2(C_B/m)]$ is increasing; (4.2) proves (4.3) for large R.

4.1 Coefficients near a distant point

Rotate so that $v_0 = Re_1$. Uniformly for $v \in B_{1/4}(v_0)$ and $\tau/4 \le t \le T$,

$$P_f(v) \le C_{H,\tau} R^{-1/2}, \qquad c_{2,f}(v) \le CR^{-2/3}. \tag{4.4}$$

For the first bound, let $g = f\mathbf{1}_{B_2(v_0)}$. Its mass is $O(R^{-2})$, and $\|g\|_3 \le CBR$ by (4.1). Splitting the Coulomb integral at ρ gives

$$\||\cdot|^{-1} * g\|_\infty \le C\|g\|_3\rho + \|g\|_1/\rho \le C\sqrt{\|g\|_1\|g\|_3} = O(R^{-1/2}).$$

The part $|w| \le R/2$ outside this ball contributes $O(R^{-1})$, and the remaining separated tail contributes $O(R^{-2})$ by energy. For the second bound take $h = (f \wedge 2)\mathbf{1}_{B_2(v_0)}$. It is bounded by two and has mass $O(R^{-2})$, so

$$\||\cdot|^{-2} * h\|_\infty \le C\rho + \|h\|_1\rho^{-2} \le C\|h\|_1^{1/3}.$$

The two far pieces are $O(R^{-2})$. This proves (4.4).

Let $C = B_{R_0}$ be the fixed core ball selected in the good-set construction of Section 2. Its mass $m \ge m_0 > 0$ and its mean $u_0 = m^{-1}\int_C wf(w)\,\mathrm{d}w$ satisfy $|u_0| \le R_0$. Recall that C contains a set E on which $f \ge \ell$ and $|E| \ge \mu_0 > 0$, with ℓ, μ_0 depending only on H. The fixed tube radius d from that construction leaves volume at least $\mu_0/2$ of E outside every line tube. In particular, for $\xi \ne 0$, outside the tube around $u_0 + \mathbb{R}\xi$, the distance to that line is at least d. Integrating over this remaining set gives

$$\int_C f(w)|\xi \times (w - u_0)|^2\,\mathrm{d}w \ge c_H|\xi|^2.$$

The core mean cancels the mixed term in

$$\begin{aligned}\int_C f|\xi \times (v - w)|^2 &= m|\xi \times (v - u_0)|^2 + \int_C f|\xi \times (w - u_0)|^2 \\ &\ge c'_H(|\xi \times v|^2 + |\xi|^2). \end{aligned} \tag{4.5}$$

Here we used $|\xi \times v|^2 \le 2|\xi \times (v - u_0)|^2 + 2R_0^2|\xi|^2$. On the core $|v - w|^3 \le C_H R^3$, so

$$\xi^T A[f\mathbf{1}_C](v)\xi \ge c_H R^{-3}(|\xi \times v|^2 + |\xi|^2).$$

If $v = Re_1 + h$, $|h| \le 1/4$, then $R^2|\xi_y|^2 \le 2|\xi \times v|^2 + 2|h|^2|\xi|^2$. Consequently the full quadratic form satisfies

$$A[f\mathbf{1}_C](v) \ge c_H R^{-1}P_y \qquad (v \in B_{1/4}(Re_1)). \tag{4.6}$$

The small radial term in (4.5) pays for the tilt of the transverse directions; a bound on the transverse block alone would not give this matrix inequality. All constants are uniform in the direction of v_0 and in time, since entropy does not increase.

4.2 Velocity dilation and the tail bound

Set

$$\mu = R^{-1/2}, \qquad F(t, w) = f(t, v_0 + \mu w), \qquad L = (4\mu)^{-1} = \sqrt{R}/4. \tag{4.7}$$

The amplitude and time are unchanged. A change of variables in the full convolution gives

$$A[F](w) = \mu^{-2}A[f](v_0 + \mu w), \quad D_w^2F = \mu^2 D_v^2 f, \quad P_F = \mu^{-2}P_f, \quad c_{2,F} = \mu^{-1}c_{2,f}. \tag{4.8}$$

In particular $F_t = A[F] : D_w^2F + F^2$, the same Coulomb equation. The cap two is unchanged because the amplitude is unchanged.

The transformed core is outside B_L for large R. Equations (4.4)–(4.6) give (3.1)–(3.2) with

$$c = c_H, \qquad \Lambda \le C_{H,\tau}R^{1/2}, \qquad \Xi \le CR^{-1/6}, \qquad \Lambda/L^2 + \Xi/L \le K_*(H, \tau), \qquad L \ge 1.$$

The rescaled mass is μ^{-3}; this does not enter Proposition 3.1. On $[a,b]=[\tau/4,T]$, (4.3) gives

$$\int_{\tau/4}^{T}\int_{B_L} F(\log_+ F+\log_+^2 F)\le C(H,\tau,T)\mu^{-3}R^{-2}\log^2 R = C(H,\tau,T)R^{-1/2}\log^2 R\longrightarrow 0. \quad (4.9)$$

Use Proposition 3.1 with $\theta=\tau/4$. Its threshold depends only on the displayed permitted constants. Choose a single $R_*=R_*(H,\tau,T)$ large enough for all the geometric conditions and (4.9) whenever $R\ge R_*$. Evaluate its conclusion at $w=0$ for every center v_0 to obtain

$$f(t,v)\le 2 \qquad (\tau/2\le t\le T,\ |v|\ge R_*). \quad (4.10)$$

This treats the entire unbounded velocity region.

4.3 The bounded velocity region

Proof of Theorem 1.1. On $[\tau/2,T]$, (4.1) and (4.10) give

$$\int f^3\le \langle R_*\rangle^3 B^3+4\int f\le C(H,\tau,T).$$

Interpolation with mass one gives a uniform L^2 bound. The potential split then bounds $A[f]$ above, and the good set gives a uniform ellipticity lower bound on B_{R_*+2}.

Write $b[f]=\operatorname{div}A[f]=\nabla a[f]$. We need

$$\|b[f]\|_6\le C\|f\|_2. \quad (4.11)$$

Here is a direct Newtonian proof. For a compact smooth source f_n, let $a_n=(4\pi|\cdot|)^{-1}*f_n$. It is smooth and $-\Delta a_n=f_n$. Outside its support $D^j a_n=O(|v|^{-1-j})$. Two integrations by parts on expanding balls therefore give $\int|D^2a_n|^2=\int(\Delta a_n)^2=\int f_n^2$. The field ∇a_n belongs to H^1, so Sobolev gives $\|\nabla a_n\|_6\le C\|f_n\|_2$. Take $f_n=\chi_n f$. The actual profile is smooth and rapidly decaying; the locally integrable gradient kernel $|z|^{-2}$ gives $\nabla a_n(v)\to b[f](v)$ by dominated convergence, and $f_n\to f$ in L^2. Fatou proves (4.11). Thus the drift is uniformly in $L_t^\infty L_v^6$ on the positive-time interval.

For completeness, perform the bounded-region iteration in the divergence equation $f_t=\operatorname{div}(A\nabla f-bf)$. On $Q=(\tau/2,T)\times B_{R_*+2}$ let $\lambda_* I\le A\le\Lambda_* I$ and $\sup_t\|b(t)\|_6\le B_*$, with permitted-data constants. Choose a compact space-time cutoff ζ vanishing at the initial time, and test by $\zeta^2 f^{p-1}$, $p\ge 2$. Put $w=f^{p/2}$ and $U=\zeta w$. After multiplication by p the principal energy is

$$\frac{4(p-1)}{p}\int\zeta^2\nabla w^T A\nabla w.$$

The diffusion cutoff term is $-4\int\zeta w\nabla\zeta^T A\nabla w$; the drift terms are $2(p-1)\int\zeta^2 wb\cdot\nabla w$ and $2p\int\zeta w^2 b\cdot\nabla\zeta$. Young's inequality spends a fixed fraction of the energy and leaves $Cp^2\int|b|^2U^2+C\int w^2|\nabla\zeta|^2$, together with the time-cutoff term. At each time,

$$\int|b|^2U^2\le B_*^2\|U\|_3^2\le CB_*^2\|U\|_2\|\nabla U\|_2.$$

A second Young inequality absorbs the gradient and leaves $Cp^4B_*^4\|U\|_2^2$. Therefore

$$\sup_t\int U^2+\int_Q|\nabla U|^2\le C[p^4(1+B_*^4)+\|\nabla\zeta\|_\infty^2+\|\zeta_t\|_\infty]\int_{\operatorname{supp}\zeta}f^p. \quad (4.12)$$

Nonnegative solutions are covered by regularizing the test power on this compact cylinder. All individual bounds used for this passage justify integration only; the constants in (4.12) are uniform.

Spatial interpolation and Sobolev imply

$$\|U\|^2_{L^{10/3}(Q)} \le C(\sup_t \|U(t)\|_2^2)^{2/5} \left(\int_Q |\nabla U|^2\right)^{3/5}.$$

Use nested cylinders with initial times $a_j = \tau - \tau 2^{-j-1}$ and radii $R_* + 1 + 2^{-j}$, cutoffs equal to one on the next cylinder, and $p_j = 2(5/3)^j$. Their derivatives are bounded by $C_\tau 4^j$. Equation (4.12) gives

$$\|f\|_{L^{p_{j+1}}(Q_{j+1})} \le [C_* p_j^4 4^j]^{1/p_j} \|f\|_{L^{p_j}(Q_j)}.$$

The product converges since $\sum j/p_j$ and $\sum \log(p_j)/p_j$ are finite. The starting space-time L^2 norm is uniformly bounded. Hence $f \le C(H, \tau, T)$ on $(\tau, T) \times B_{R_*+1}$, first almost everywhere and then pointwise by continuity, including the time endpoints. Together with (4.10) this proves the global bound in Theorem 1.1. □

AI-use statement

OpenAI Codex assisted with proof development, source checking and manuscript preparation. The author is responsible for the mathematical claims and references.

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